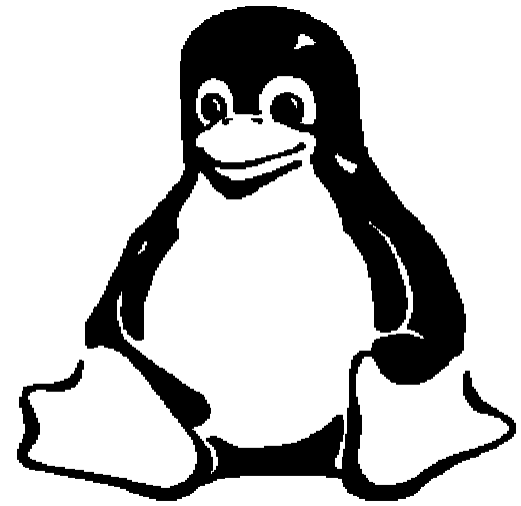


An introduction to Linux command shell



lab2

What is a command shell?

- ✓ *A program that interprets commands*
- ✓ *Allows a user to execute commands by typing them manually at a terminal, or automatically in programs called shell scripts.*
- ✓ *A shell is not an operating system. It is a way to interface with the operating system and run commands.*

What is *BASH*?

- ✓ *BASH = Bourne Again SHell*
- ✓ *Bash is a shell written as a free replacement to the standard Bourne Shell (/bin/sh) originally written by Steve Bourne for UNIX systems.*
- ✓ *It has all of the features of the original Bourne Shell, plus additions that make it easier to program with and use from the command line.*
- ✓ *Since it is Free Software, it has been adopted as the default shell on most Linux systems.*

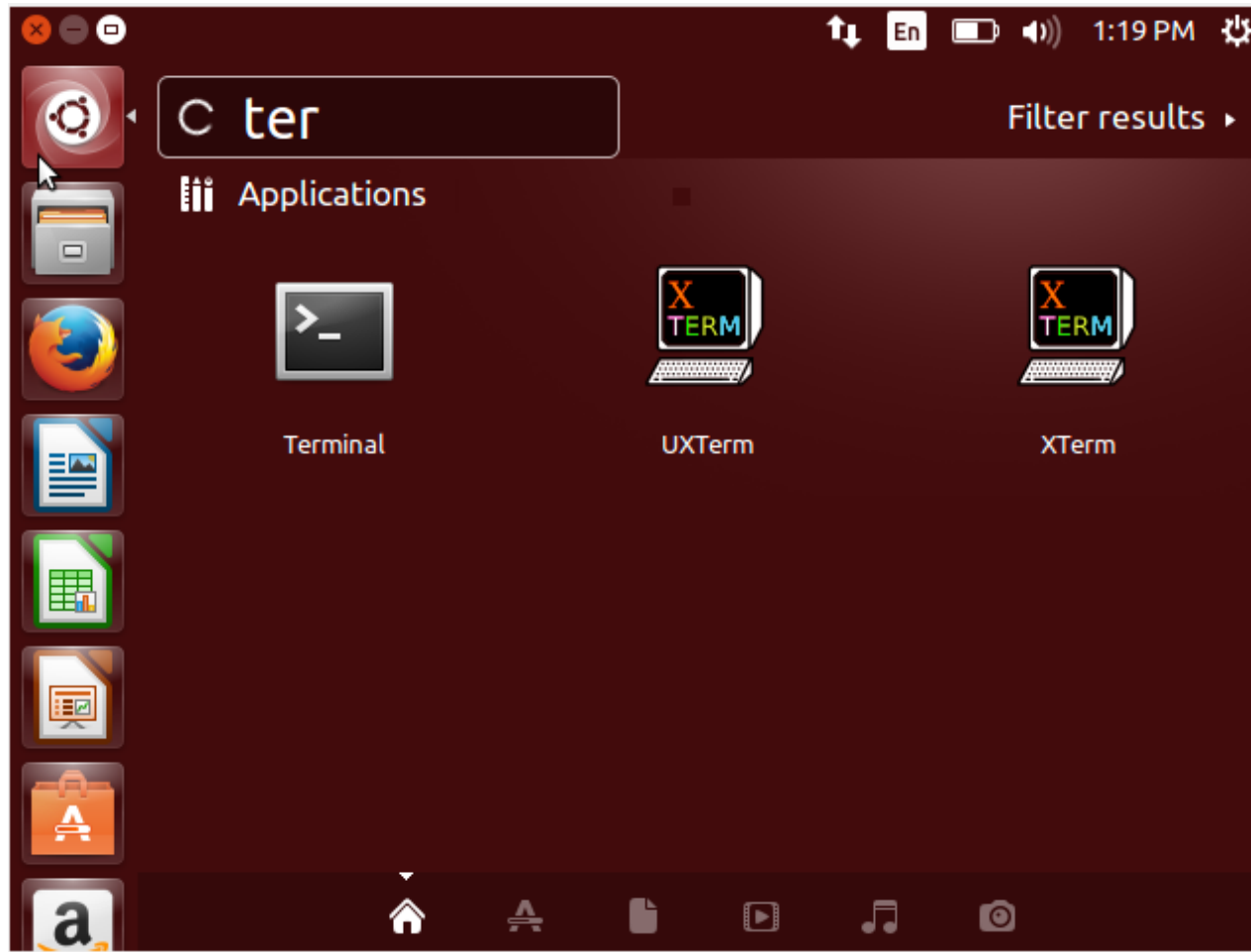
How is BASH different from the DOS command prompt?

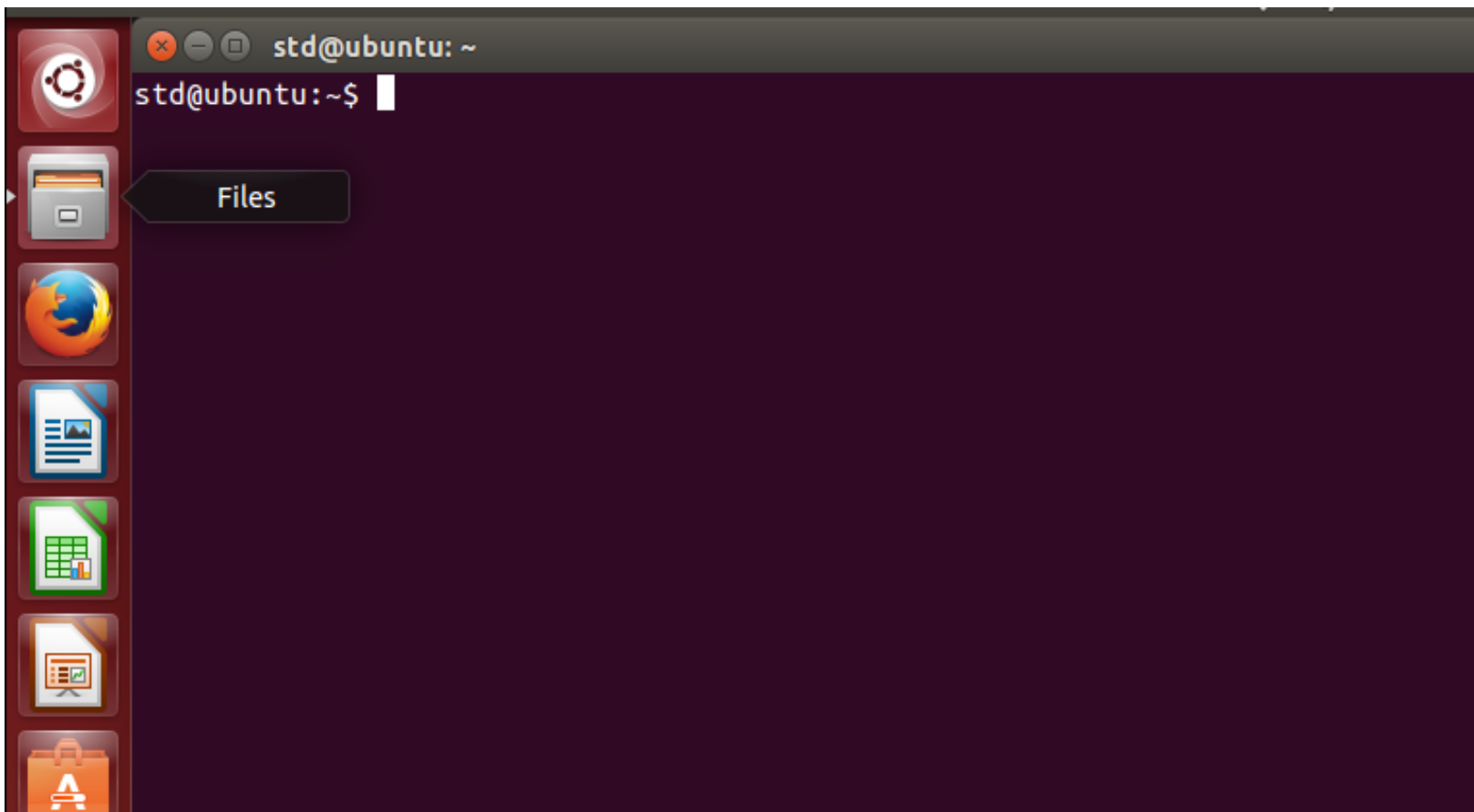
- ✓ **Case Sensitivity:** In Linux/UNIX, commands and filenames are case sensitive, meaning
- ✓ that typing “EXIT” instead of the proper “exit” is a mistake.
- ✓ **“\” vs. “/”:** In DOS, the forward-slash “/” is the command argument delimiter, while the backslash “\” is a directory separator. In Linux/UNIX, the “/” is the directory separator, and the “\” is an escape character. More about these special characters in a minute!

✓ **Filenames:** The DOS world uses the “eight dot three” filename convention, meaning that all files followed a format that allowed up to 8 characters in the filename, followed by a period (“dot”), followed by an option extension, up to 3 characters long (e.g. `FILENAME.TXT`). In UNIX/Linux, there is no such thing as a file extension. Periods can be placed at any part of the filename, and “extensions” may be interpreted differently by all programs, or not at all.

Command Prompt (Windows) $\longrightarrow$ Terminal (Linux)

- **Alt + Ctrl + T** to open the terminal or :





Std : user name

Ubuntu : machine name (computer name)

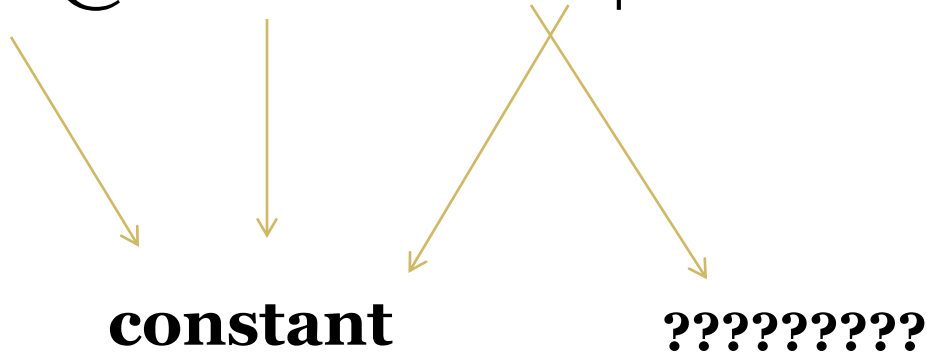
~ : /home/std

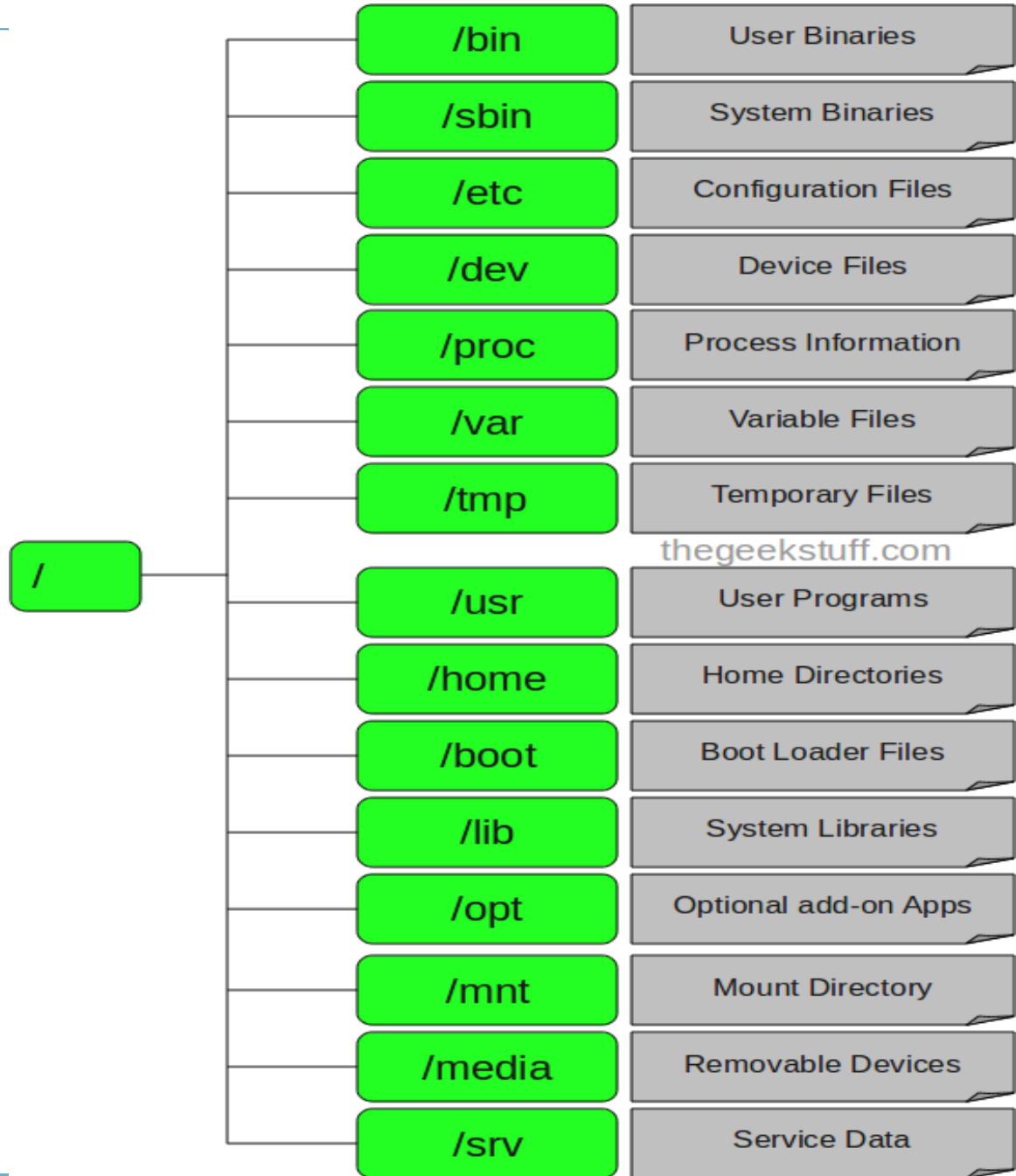
Home Folder ?

Users In Linux :

- Normal User \$
 - Root #
-

std@ubuntu: ~ \$





Directory commands

Command	Function
ls	List all files in the current directory, in column format.
cd	“Change Directory”. When typed all by itself, it returns you to your home directory.
pwd	“Print Working Directory”. Shows the current location in the directory tree.
mkdir	Make Directory.
rmdir	Remove Directory.

ls Command

Ls Command	Function
ls directory	List the files in the specified directory. Example: <code>ls /var/log</code>
ls -l	List files in “long” format, one file per line. This also shows you additional info about the file, such as ownership, permissions, date, and size.
ls -a	List all files, including “hidden” files. Hidden files are those files that begin with a “.”, e.g. The <code>.bash_history</code> file in your home directory.
ls /usr/bin/d*	List all files whose names begin with the letter “d” in the <code>/usr/bin</code> directory.
<u>ls</u> -s	List all files and size.

ls command

std@ubuntu:~\$ ls

Desktop	Downloads	Music	Public	Videos
Documents	examples.desktop	Pictures	Templates	

ls command

std@ubuntu:~\$ ls -l

```
total 44
drwxr-xr-x 2 std std 4096 Feb 26 04:23 Desktop
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Documents
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Downloads
-rw-r--r-- 1 std std 8980 Feb 25 05:19 examples.desktop
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Music
drwxr-xr-x 2 std std 4096 Feb 26 22:55 Pictures
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Public
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Templates
drwxr-xr-x 2 std std 4096 Feb 25 05:27 Videos
```

ls command

std@ubuntu:~\$ ls -l

total 44

drw-r-xr-x 2 (directory)

-rw-r--r- 1 (file)

std std (user name / group name)

4096 (size)

ls command

std@ubuntu:~\$ ls -a

```
.          .dmrc          Pictures
..         Documents    .profile
.bash_history Downloads      Public
.bash_logout examples.desktop Templates
.bashrc     .gconf          Videos
.cache      .ICEauthority  .Xauthority
.config     .local         .xsession-errors
Desktop     Music          .xsession-errors.old
std@ubuntu:~$
```

- . Current directory
- .. parent directory

ls command

std@ubuntu:~\$ ls -al

or ls -la

or ls -a -l

```
std@ubuntu:~$ ls -al
```

```
total 104
```

```
drwxr-xr-x 14 std std 4096 Feb 26 02:22 .
```

```
drwxr-xr-x  3 root root 4096 Feb 25 05:19 ..
```

```
-rw----- 1 std std   5 Feb 25 05:28 .bash_history
```

```
-rw-r--r-- 1 std std  220 Feb 25 05:19 .bash_logout
```

```
-rw-r--r-- 1 std std 3760 Feb 25 05:19 .bashrc
```

```
drwx----- 13 std std 4096 Feb 26 04:20 .cache
```

```
drwx----- 16 std std 4096 Feb 26 04:21 .config
```

```
drwxr-xr-x  2 std std 4096 Feb 26 04:23 Desktop
```

```
-rw-r--r-- 1 std std  25 Feb 25 05:27 .dmrc
```

ls command

std@ubuntu:~\$ ls /home/std/

Desktop	Downloads	Music	Public	Videos
Documents	examples.desktop	Pictures	Templates	

ls command

std@ubuntu:/\$ ls -s

```
total 96
```

```
4 bin    4 home      4 media  0 run    4 usr
4 boot   0 initrd.img  4 mnt    12 sbin   4 var
4 cdrom  4 lib        4 opt    4 srv    0 vmlinuz
0 dev    4 lib64      0 proc   0 sys
12 etc   16 lost+found 4 root   4 tmp
```

```
std@ubuntu:/$
```

ls command

std@ubuntu:/\$ ls /usr/bin/d*

```
/usr/bin/dbus-launch  
/usr/bin/dbus-monitor  
/usr/bin/dbus-run-session  
/usr/bin/dbus-send  
/usr/bin/dbwrap_tool  
/usr/bin/dc  
/usr/bin/dconf  
/usr/bin/deallocvt  
/usr/bin/debconf  
/usr/bin/debconf-apt-progress  
/usr/bin/debconf-communicate  
/usr/bin/debconf-copydb
```

ls command

std@ubuntu:/\$ ls /usr/bin/d*

```
/usr/bin/dbus-launch  
/usr/bin/dbus-monitor  
/usr/bin/dbus-run-session  
/usr/bin/dbus-send  
/usr/bin/dbwrap_tool  
/usr/bin/dc  
/usr/bin/dconf  
/usr/bin/deallocvt  
/usr/bin/debconf  
/usr/bin/debconf-apt-progress  
/usr/bin/debconf-communicate  
/usr/bin/debconf-copydb
```

cd Command	Function
cd directory	Change into the specified directory name. Example: cd /usr/src/linux
cd ..	Return to previous directory(Move up one directory). For example, if you are in /home/vic and you type “cd ..”, you will end up in /home.
cd ~	“~” is an alias for your home directory. It can be used as a shortcut to your “home”, or other directories relative to your home.
cd -	Return to previous location. An easy way to get back to your (previous location)!
cd /	To switch to the root directory.

cd command

std@ubuntu:~\$ cd ..

```
std@ubuntu:/home$ cd ..  
std@ubuntu:/$
```

```
std@ubuntu:~$ cd ..  
std@ubuntu:/home$ cd -  
/home/std  
std@ubuntu:~$
```

cd command

std@ubuntu:/\$ cd ~

```
std@ubuntu:~$
```

cd command

std@ubuntu:/\$ cd -

```
/
```

std@ubuntu:~\$ cd -

```
/home
```

cd command

std@ubuntu:/\$ cd home

```
std@ubuntu:/home$
```

cd command

std@ubuntu:~\$ cd /

```
std@ubuntu:/$
```

cd command

std@ubuntu:~\$ cd /

```
std@ubuntu:/$
```

```
std@ubuntu:~$ cd /  
std@ubuntu:/$ cd home  
std@ubuntu:/home$ cd std  
std@ubuntu:~$ cd /  
std@ubuntu:/$ cd ~  
std@ubuntu:~$ cd /  
std@ubuntu:/$
```

Pwd command

std@ubuntu:~\$ pwd

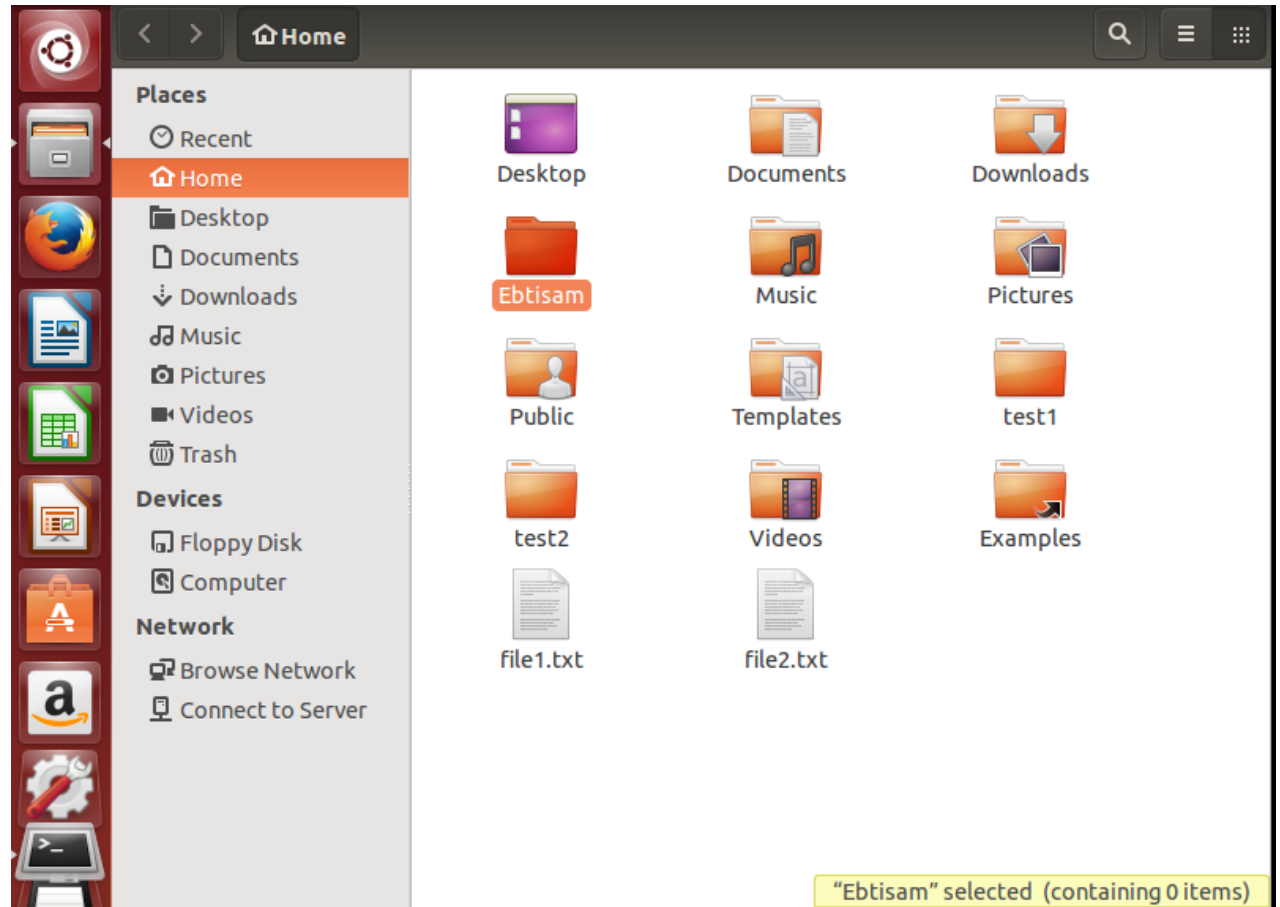
```
std@ubuntu:~$ pwd
/home/std
std@ubuntu:~$ cd /
std@ubuntu:/$ pwd
/
std@ubuntu:/$
```

mkdir command

Command	Function
mkdir dir	A directory is created in the given name under the current directory. Example: mkdir test
mkdir dir1 dir2	A number of sub-directories can be created under one stroke.
mkdir -p	Creates all the required parent directories.

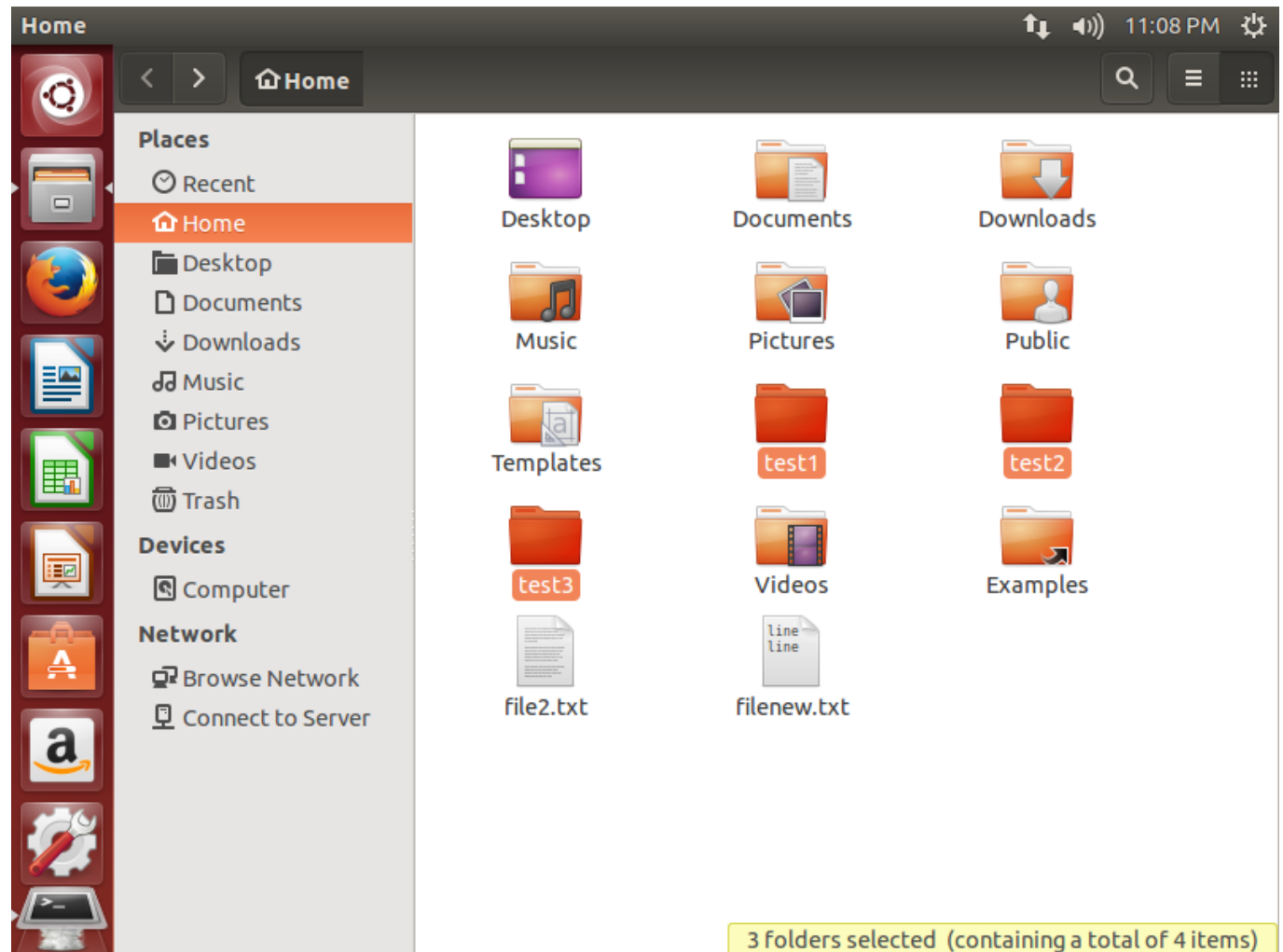
mkdir command

`std@ubuntu:~$ mkdir Ebtisam`



mkdir command

`std@ubuntu:~$ mkdir test1 test2 test3`



mkdir command

```
std@ubuntu:~$ mkdir -p ee
```

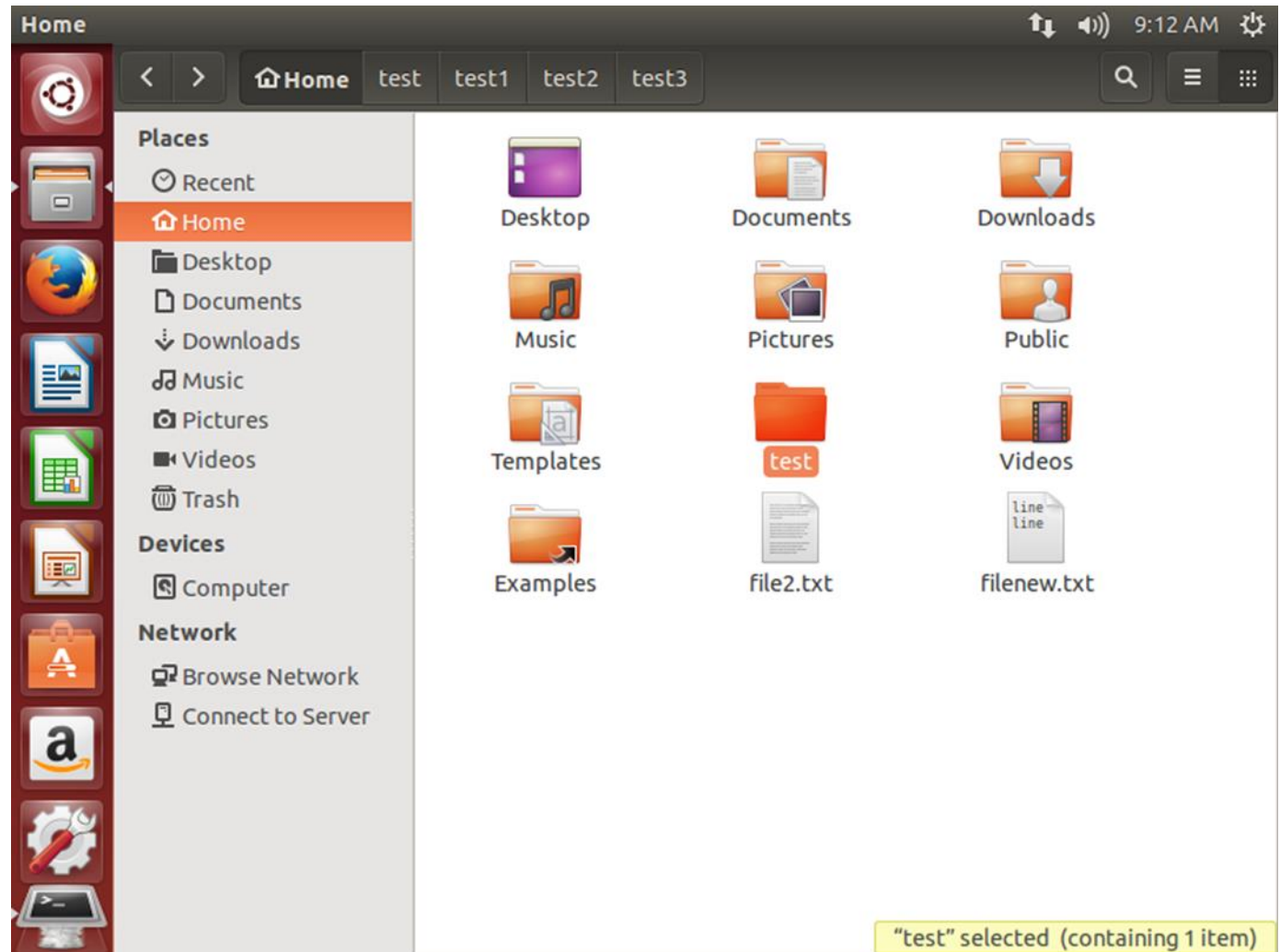
```
std@ubuntu:~$ cd ee
```

```
std@ubuntu:~/ee$ mkdir E e
```

```
std@ubuntu:~/ee$
```

mkdir command

std@ubuntu:~\$ mkdir -p test/test1/test2/test3

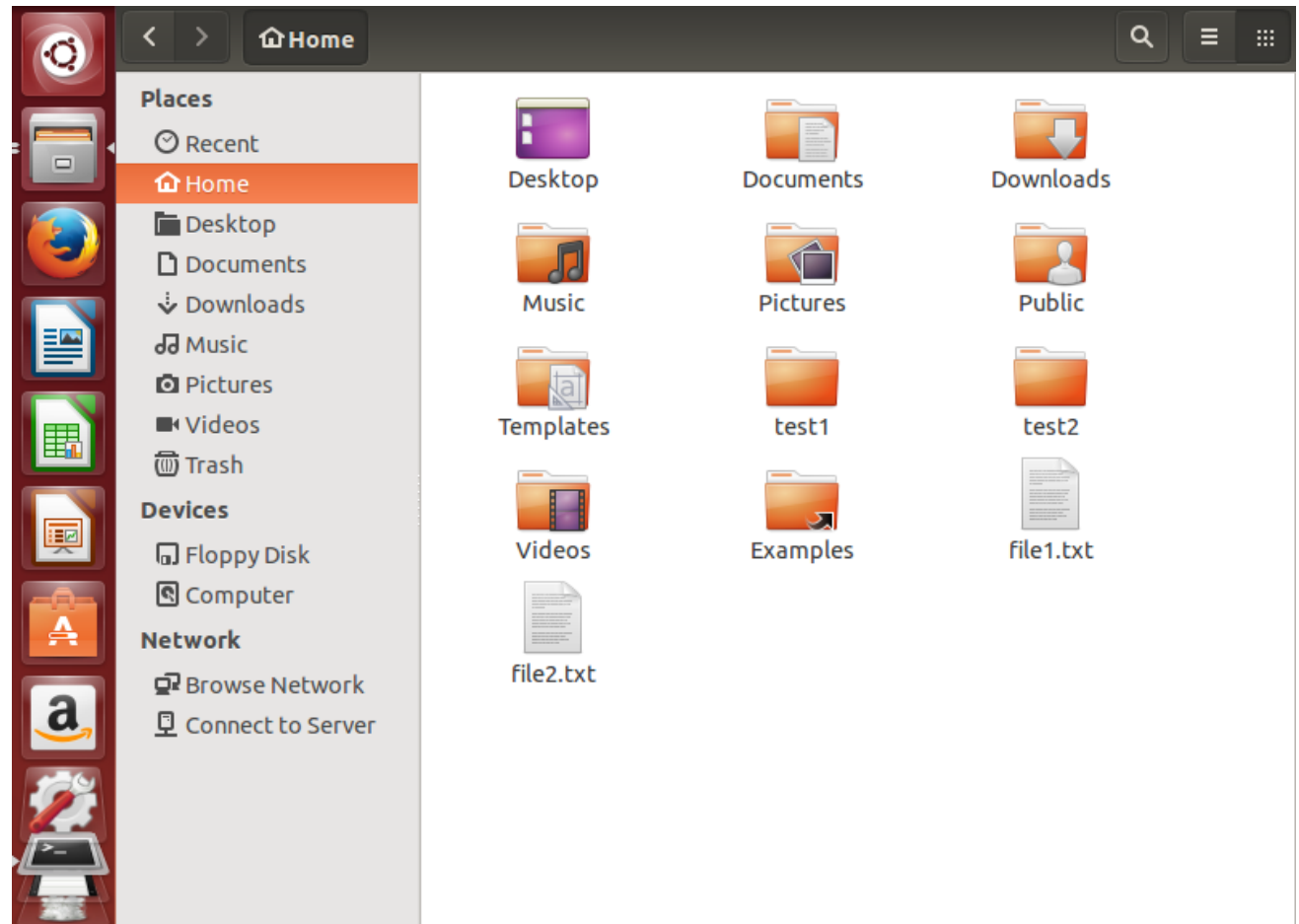


rmmdir command

Command	Function
rmmdir dir	Removes an empty directory.
rm -r dir	remove a directory and its contents.

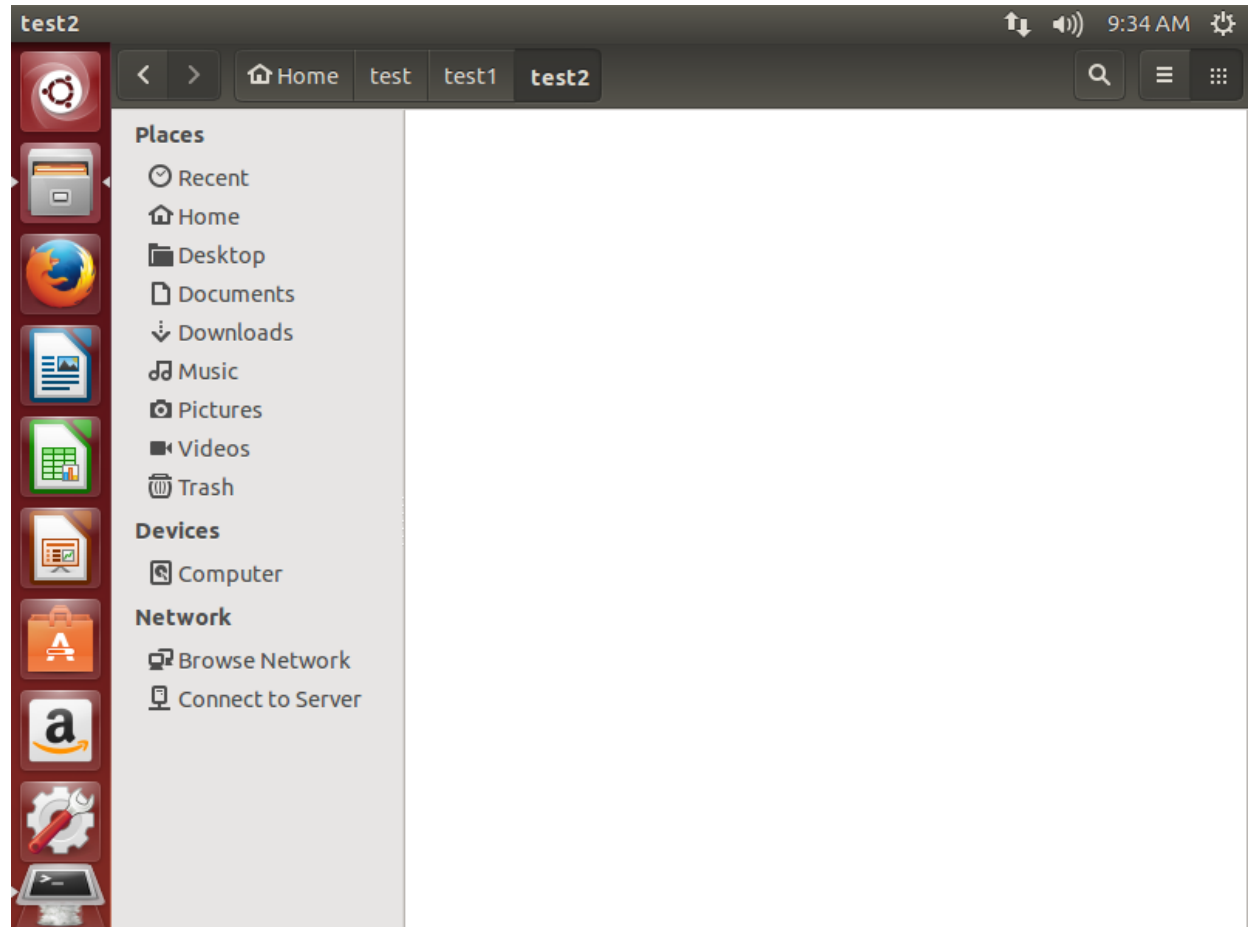
rmmdir command

std@ubuntu:~\$ rmmdir Ebtisam



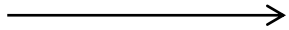
rmmdir command

std@ubuntu:~\$ rm -r test/test1/test2/test3



rm-r command

```
std@ubuntu:~$ rm -r test
```



remove a directory and its contents.

rm-r command

```
std@ubuntu:~/test$ rm -r *
```



Deletes all files and sub-directories

rm-r command

```
std@ubuntu:~/test$ rm *
```



To delete all the **files** in the directory.

mv & cp Commands

Command	Function
mv	Moves a directory to a new location, or renames it.
cp -r	Copies a directory from one location to another.

mv command

```
std@ubuntu:~$ mv test test1
```



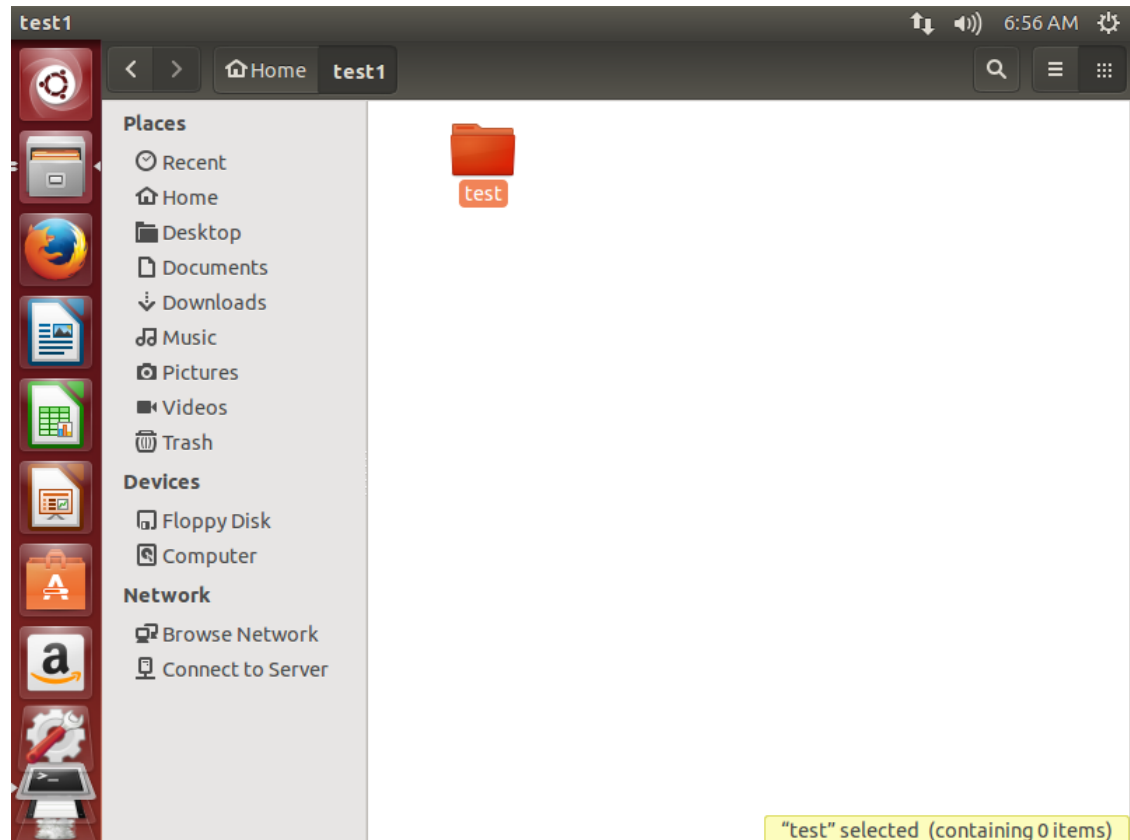
= rename

mv command

std@ubuntu:~\$ mv test test1/test



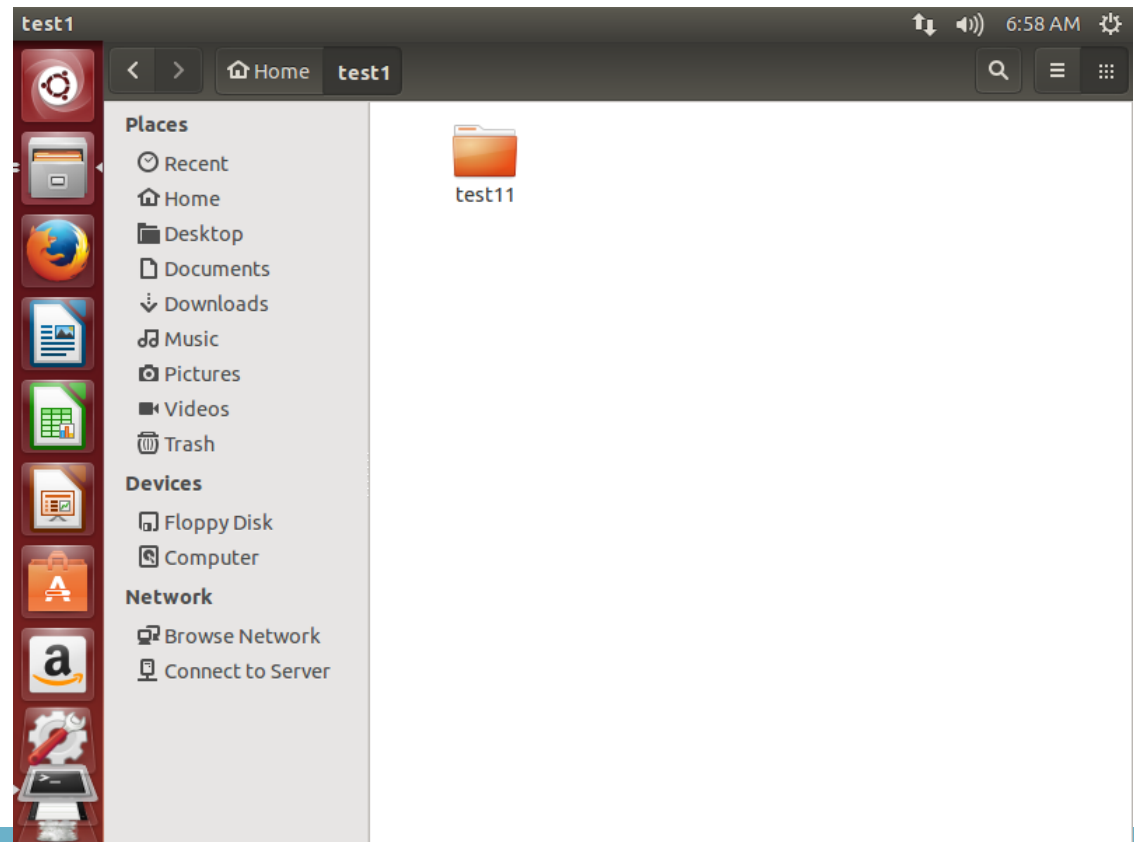
= move (with same name)



mv command

`std@ubuntu:~$ mv test test1/test11`

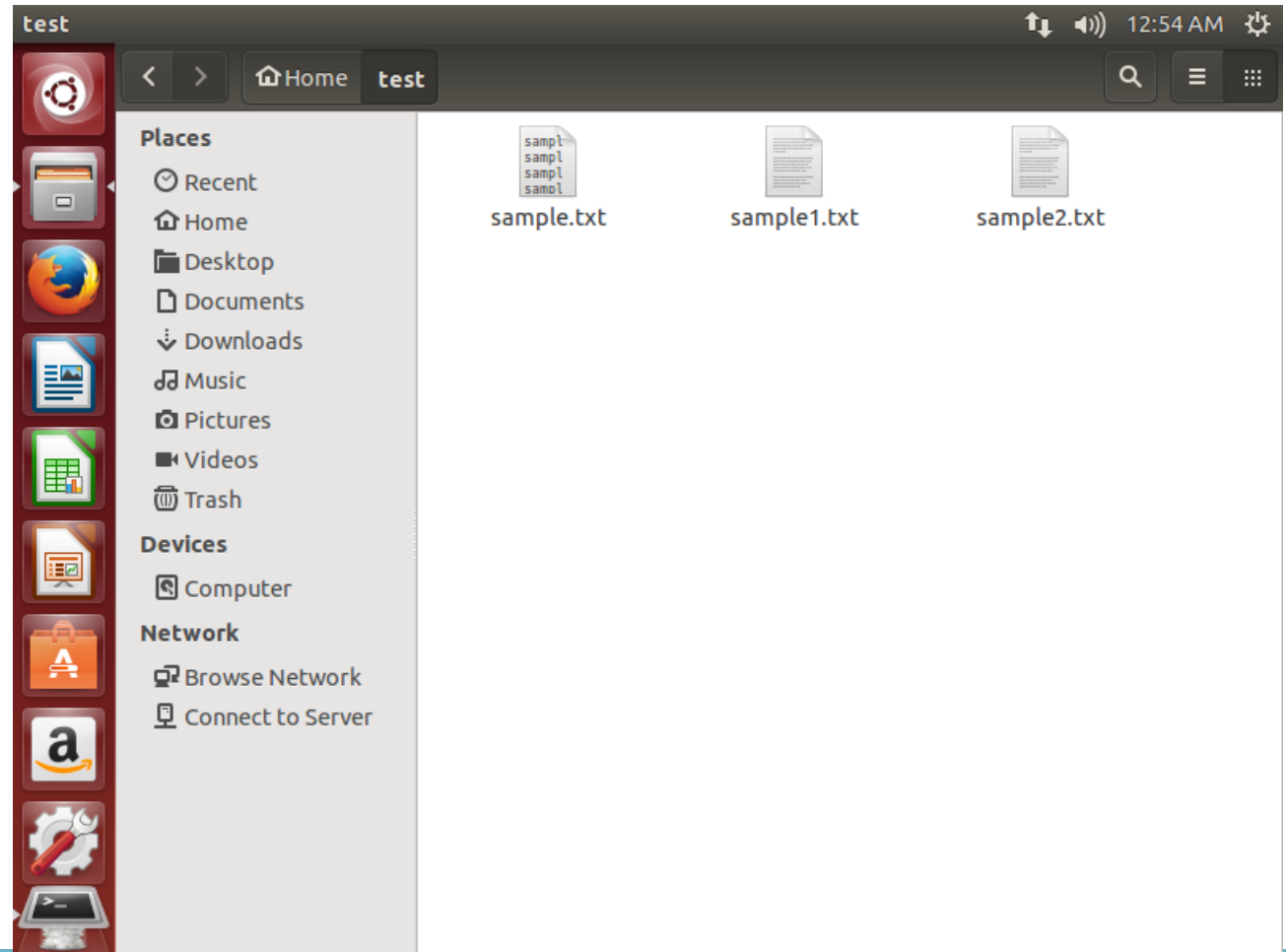
-----> = move (with different name)



mv command

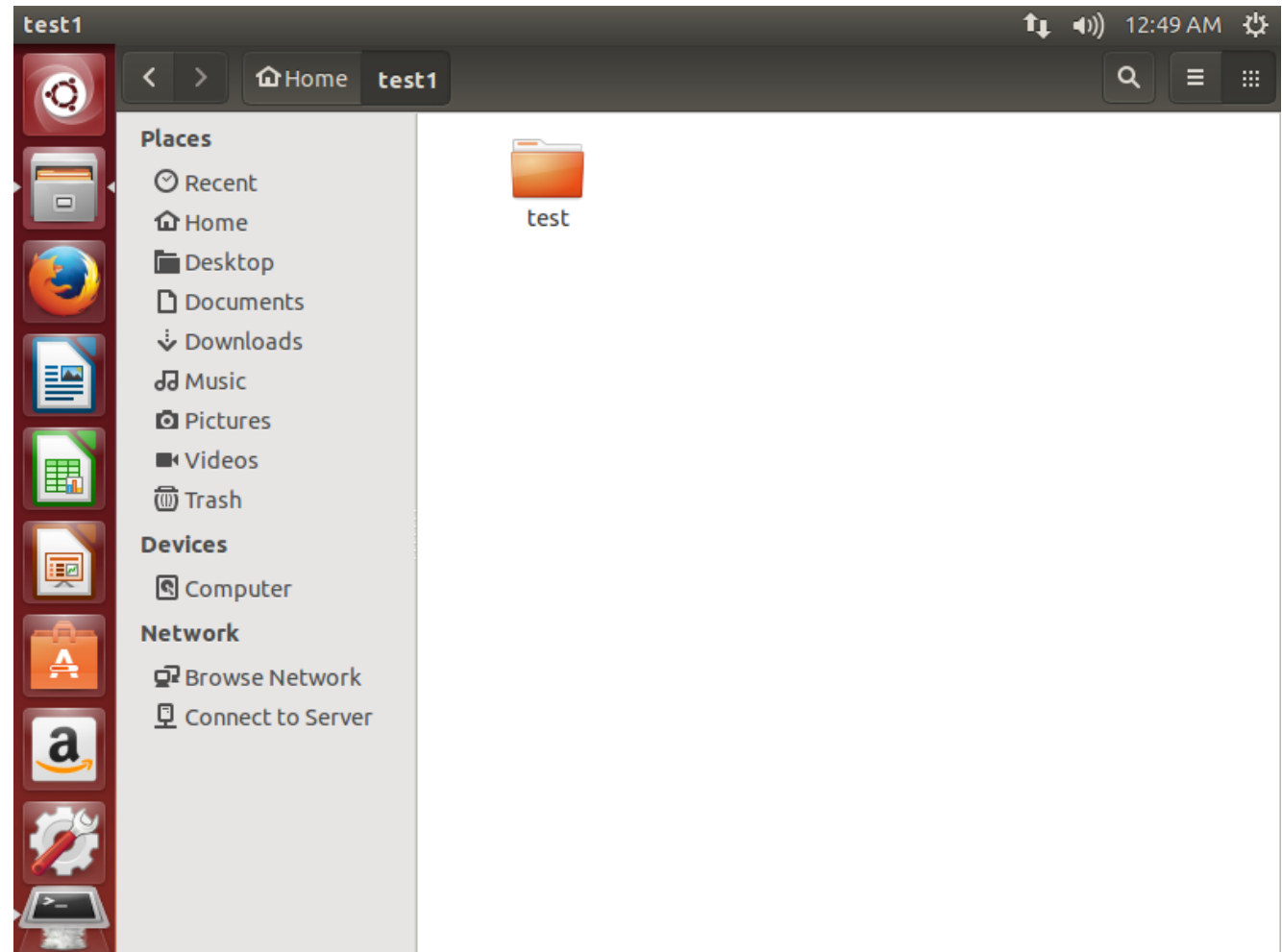
```
std@ubuntu:~$ mv sample.txt sample1.txt  
sample2.txt      test
```

To move a group of files to a directory.



cp -r command

std@ubuntu:~\$ cp -r test test1



General Commands

General Command	Function
history	Show commands listed in the shell history (last n).
clear	Clear the terminal screen.
cal	Calendar. ex: cal 2013 shows a calendar of the year 2013

General Command	Function
cal	ex: cal 12 2013 std@ubuntu:~\$ cal 12 2013 December 2013 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 std@ubuntu:~\$ cal -m feb February 2019 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28
T / Ebtisam ALselwi	48

General commands

Command	Function
date	Print or set the system date and time. <code>std@ubuntu:~\$ date</code> Tue Mar 5 22:06:25 PST 2019 <code>std@ubuntu:~\$ date +%D</code> 03/05/19 Displays date only. <code>std@ubuntu:~\$ date +%T</code> 22:13:58 Displays time only. <code>std@ubuntu:~\$ date +%Y</code> 2019 Displays the year part of date. <code>std@ubuntu:~\$ date +%H</code> 22 Displays the hour part of time.

General commands

Command	Function
echo "txt"	<code>std@ubuntu:~\$ echo "Ebtisam"</code> Ebtisam Displays the given text on the screen
echo \$HOME	<code>std@ubuntu:~\$ echo \$HOME</code> Displays the user's home directory. /home/std
Exit	Exit from a process. If shell is the only process then logs out

End